

Unit 06: Direction Sense Test

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Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- Understand Distance Related Questions
- Understand Left Right Movement

Introduction

The questions based on directions require the candidates to identifying the direction of an individual or shadow from a set of statements. We should know of some of the important points which helps to solve the questions with ease.

6.1 General Directions

In general, there are four main directions i.e. North, South, East and West. Apart from these four, there are four additional directions derived from the main ones. They are called North-East, North-West, South-East and South-West. A chart is given below for reference.

6.2 Casting of Shadows

A lot of times questions in CAT are asked regarding casting of shadows. It is important and useful to know the below concepts to understand

If a man faces a rising sun, his shadow will always be in the west.

If a man faces north, his shadow will be on his right during sunrise and on his left during sunset.

Similarly, if a man faces south, his shadow will be on his left during sunrise and on his right during sunset.

In mid-noon, no shadows are seen as the sun's rays are vertically downwards.

6.3 Distance Related Questions

In most of the direction and distance questions, it is required to calculate a certain parameter from the question statement. The questions can be related to the total distance walked, shortest path, the distance between two entities, etc.

While solving the distance related question, one must be thorough with the Pythagorean Theorem to be able to solve most of the questions. The Pythagorean Theorem is used to calculate the shortest path traveled, the minimum distance between two points, etc. A couple of examples and cat questions are given below to illustrate the direction and distance questions better.

6.4 Left Right Movement

A person facing north, on taking left will face towards west and on taking the right turn towards east.

A person facing west, on taking left will face towards south and on taking right turn towards north.

A person facing east, on taking left will face towards north and on taking the right turn towards south.

A person facing south, on taking left will face towards east and on taking the right turn towards west.



Example 1: Airplanes A, B, C and D started flight towards east. After flying 125 kms planes A and D flew towards right while planes B and C flew towards left. After 115 km, planes B and C flew towards their left while planes A and D also turned towards their left. In which directions are the airplanes A, B, D, C respectively flying now?

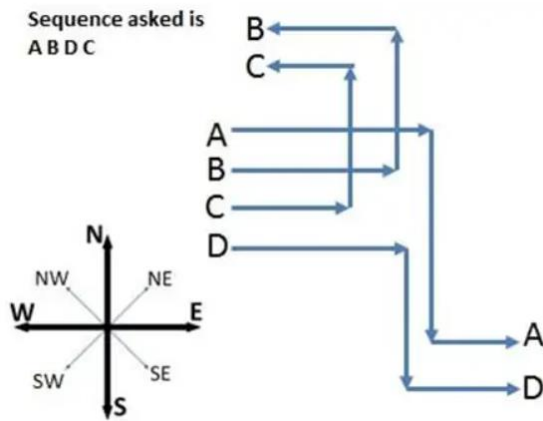
1. North, South, East, West
2. East, West, West, East
3. East, West, East, West
4. South, North, North, South

Let us look at A and D first. After flying for 125 kms, A and D would still be in east direction. Now, they take a right turn which makes their direction as south. From south, if you take a left, you will again come in the east direction. So planes A and D are in east direction.

Now, let us look at planes B and C. After flying for 125 kms, B and C would still be in east direction.

From east direction, they take a left turn which means now they are in north. From north, they again take a left which makes them land in the west.

So correct answer would be: East, west, East, West (Option C)



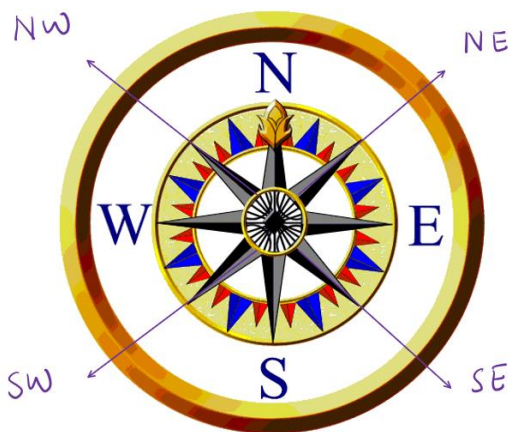
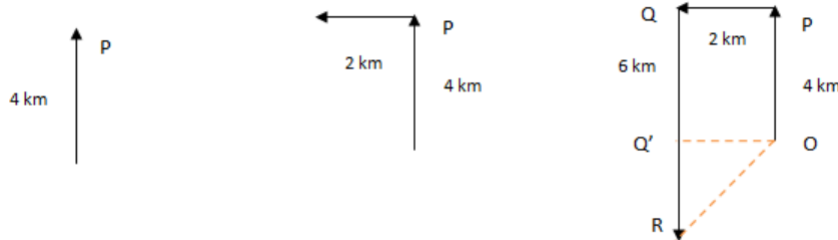
Example 2: A person starts walking towards North for about 4 km and reaches a point P. From point P, he takes a left turn and walks for 2 km to reach a point Q. From point Q, he again takes a left turn and walks for 6 km and reaches a point R. How far and in which direction is the point R from the starting point?

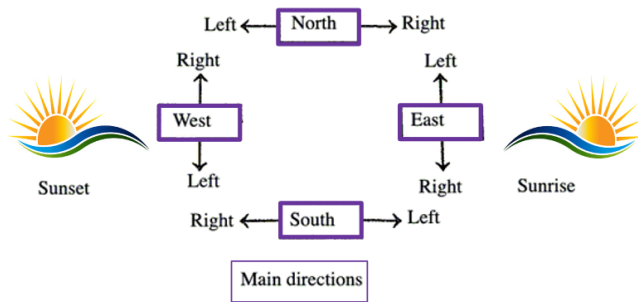
Solution: Let us first draw a rough sketch of the above reasoning direction problem.

Joining the starting point O to the endpoint R and drawing a perpendicular on QR at Q'.

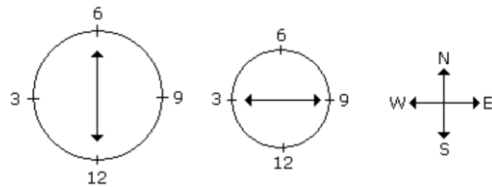
In a right angled triangle OQ'R, We have $OQ' = PQ = 2$ km, and $Q'R = QR - QQ' = 6 - 4 = 2$ km.

So, $OR^2 = 2^2 + 2^2 = 4 + 4 = 8$. or, $OR = \sqrt{8} = 2\sqrt{2}$ km. R is in the South-West direction from the starting point O.





Example 3:- Rahul put his timepiece on the table in such a way that at 6 P.M. hour hand points to North. In which direction the minute hand will point at 9.15 P.M. ?



At 9.15 P.M., the minute hand will point towards west.



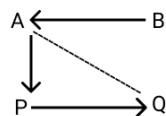
Example 4:- Town A is to the West of Town B. Town P is to the South of Town A. Town Q is to the East of Town P. Then Town Q is towards which direction of Town A?

(a) South

(b) South West

(c) South East

(d) North



Solution- (c) South East



Example 5:- Siva starting from his house, goes 5 km in the East, then he turns to his left and goes 4 km.

Finally, he turns to his left and goes 5 km. Now how far is he from his house and in what direction?

1. If North-east becomes West and South-east becomes North then what will West

becomes?

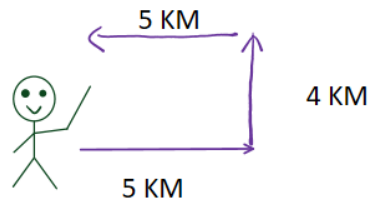
A. South-east

B. North-east

C. South

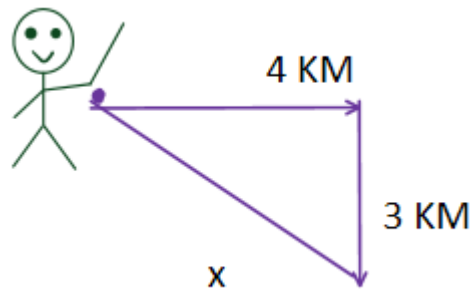
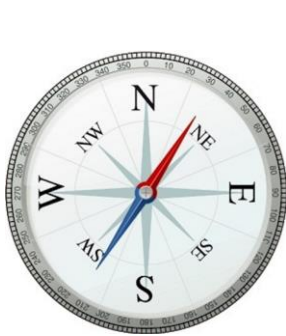
D. North-west

E. None of these



Suresh starting from his house, goes 4 km in the East, then he turns to his right and goes 3 km.

What minimum distance will be covered by him to come back to his house?

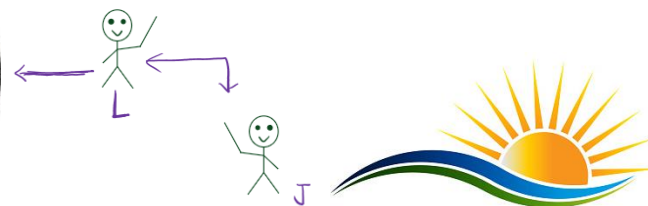
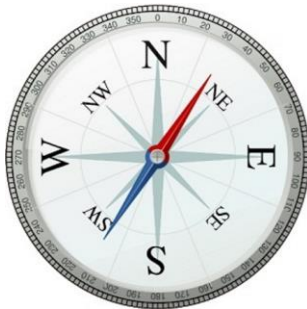


$$X^2 = 4^2 + 3^2 = 16 + 9 = 25$$

$$X = 5.$$



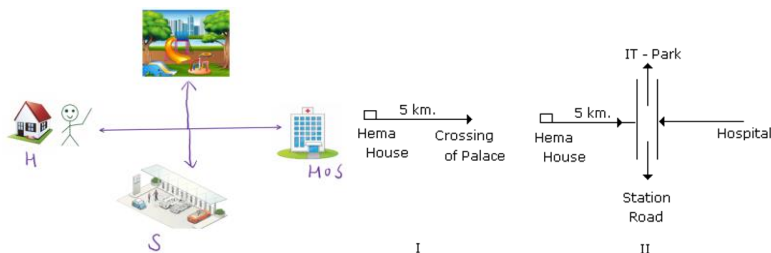
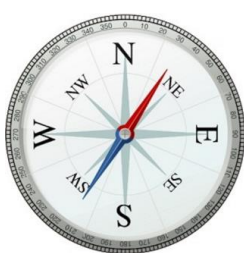
Example 6:- One morning after sunrise Juhi while going to school met Lalli at Boring road crossing. Lalli's shadow was exactly to the right of Juhi. If they were face to face, which direction was Juhi facing?



Hema starting from her house walked 5 km to reach the crossing of Palace.

In which direction she was going, a road opposite to this direction goes to Hospital.

The road to the right goes to station. If the road which goes to station is just opposite to the road which IT-Park, then in which direction to Hema is the road which goes to IT-Park?



Analytical Skills-I

It is clear that the road which goes to IT-Park is left to Hema.



Example 7 – One morning after sunrise, Mahesh was standing facing a pole. The shadow of the pole fell exactly to his right. To which direction was he facing?

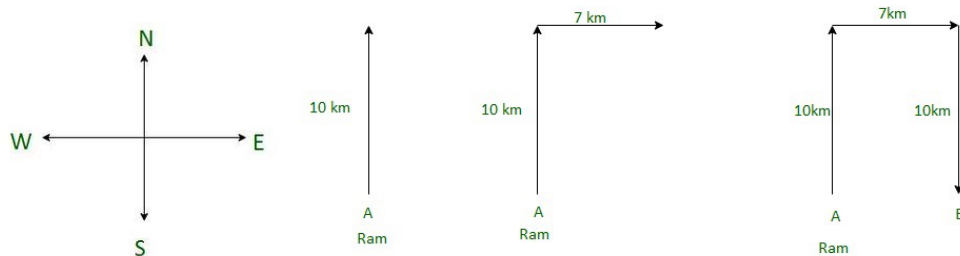
Solution –

The sun rises in the East (E) in the morning. As the shadow of Mahesh falls to his right.

So he must be facing South. Hence, the answer is the south.



Example 8- Ram starts from a point A walks 10 km north, then turns right and walks for 7 km, then turns right again and walks for another 10 km. And reaches point B. How far is Ram from the starting point?



Example 9:

One evening, two friends Riya and Priya were talking to each other, with their backs towards each other, sitting in a park. If Riya's shadow was exactly to the left of her, then which direction was Priya facing?

Solution:

Riya's shadow fell to her left i.e. towards East (as it was evening). So, Riya was facing "South". As Priya had her back towards Riya, hence, Priya was facing "North". North is the correct answer.

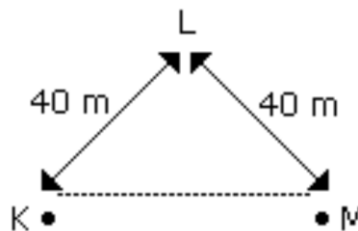


Example 10:

K is 40 m South-West of L. If M is 40 m South-East of L, then M is in which direction of K?

Solution:

M is in the East of K as shown in the following diagram. East is the correct answer.

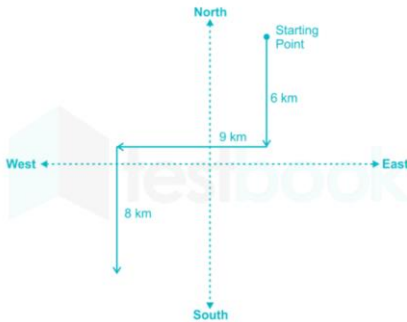


Example 11:- A man walks 6 km towards the north, then turns towards his left and walks for 4 km. He again turns left and walks for 6 km. At this point he turns to his right and walks for 6 km. How many km and in what direction is he from the starting point?

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- A. 10 km, West
- B. 6 km, South
- C. 4 km, South
- D. 8 km, West
- E. 10 km, East

Solution:- A. 10 km, West



EXAMPLE 12:- If North-east becomes West and South-east becomes North then what will West

becomes?

- A. South-east
- B. North-east
- C. South
- D. North-west
- E. None of these



EXAMPLE 13:-While facing East you turn to your left and walk 10 yards; then turn to your left and walk 10 yards: and now you turn 45° towards your right and go straight to cover 25 yards. Now, in which direction are you from your starting point ?

- A. North-East
- B. South-West
- C. East
- D. North-West
- E. None of these

Solution:- D. North-West

Let be the initial position of the man. He initially faced East and then turned his left in the direction of North and walked 10 yards. Then, turned his left in the direction of West and walked 10 yards.

Now, he turned 45° to his right and walked 50 yards straight in the same direction. Now, the direction of the man with respect to his starting point is North - West.



EXAMPLE 14:-A direction pole was situated on the crossing. Due to an accident the pole turned

in such a manner that the pointer which was showing East, started showing South.

One traveler went to the wrong direction thinking it to be West. In what direction actually he was travelling ?

Analytical Skills-I

- A. South
- B. East
- C. West
- D. North
- E. None of these

Solution:- D. (North We can see that according to graph Due to an accident, the pointer showing East, started showing South. It means, the pole has been rotated through 90° clockwise. So, when she was travelling towards West, actually it was North.)



EXAMPLE 15:-My friend is facing east. She turned 120° in the clockwise direction and then 165°

in the anti-clockwise direction. Which direction is she facing now?

- A. East.
- B. North-east.
- C. North.
- D. South-west.
- E. None of these

Answer:- B. North-east.



EXAMPLE 16:-Five friends are playing a game. Nitin is facing South. He follows instructions given by his friends.

- (i) When Kimi touch him, Nitin walks 50 metre.
- (ii) When Neha touch him, Nitin turns right and walks 40 metre.
- (iii) When Annu touch him, Nitin turns left and walks 80 metre.
- (iv) When Punit touch him, Nitin walks 70 metre in the direction opposite to which he faces.

If Kimi, Annu, Punit and Neha touch him in that order, how far does Nitin reach from his starting point?

- A. 60 metre.
- B. 72 metre.
- C. 80 metre.
- D. 75 metre.
- E. None of these

Solution:- B. 72 metre.



EXAMPLE 17:- Read the situations given below to answer the questions:

Nine cars P, Q, R, S, T, U, V, W and X are parked such that:

- (1) R is 4 km east of Q.
- (2) P is 2 km north of Q.
- (3) W is 4 km south of P.

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- (4) V is 2 km west of W.
 (5) S is 6 km east of V.
 (6) U is 4 km north of V.
 (7) X is parked in the middle of Q and R.
 (8) T is parked in the middle of W and S.

What is the distance between W and S?

- A. 6 km.
 B. 4 km.
 C. 3 km.
 D. 2 km.
 E. 5 km

Solution:- B. 4 km.



EXAMPLE 18:-What is the direction of Q with respect to W?

- A. East
 B. West
 C. South
 D. North
 E. Canno

Solution:- D. North



EXAMPLE 19:- What is the distance between U and W?

- A. 20 km
 B. $\sqrt{28}$ km
 C. $\sqrt{20}$ km
 D. $3\frac{4}{5}$ km
 E. Cannot be determined

Solution:- C. $\sqrt{20}$ km



EXAMPLE 20:- Which car is parked between V and T?

- A. S
 B. Q
 C. W
 D. T
 E. Ut be determined

Solution:- C. W



EXAMPLE 21:- A road network has parallel and perpendicular roads running north-south or east-west only. Junctions/ Intersections on this road network are marked as A, B, C, D.... All junctions are at exactly half a kilometer distance from each other. The following is known about junctions A, B, C, H and X.

'A' is east of 'B' and west of 'C', 'H' is southwest of 'C' and southeast of B. 'B'

is southeast of 'X'. Which junctions are the farthest south and the farthest east?

Analytical Skills-I

A. H, B

B. H, C

C. C, H

D. B, H

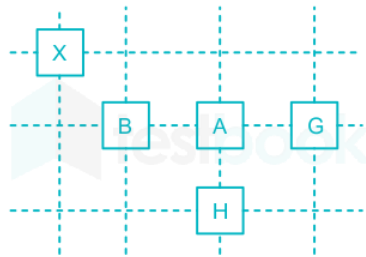
E. B, A

Solution:- B. H, C

A road network has parallel and perpendicular roads running north-south or east-west only. Junctions/ Intersections on this road network are marked as A, B, C, D.... All junctions are at exactly half a kilometer distance from each other. The following is known about junctions A, B, C, H and X.

'A' is east of 'B' and west of 'C', 'H' is southwest of 'C' and southeast of B. 'B'

is southeast of 'X'. Which junctions are the farthest south and the farthest east? Hence farthest south is H and Farthest East is C. Hence option 2 is correct answer.



DIRECTIONS ( EXAMPLE 22:-) : Ashini, Bindiya and Sanika are standing at the corners of an equilateral triangle drawn in an open farm. Study the questions given below which are based on the following diagram and answer them by marking the appropriate choice from the choices.

From the positions shown in the figure, if Ashini, Bindiya and Sanika run along the sides in clockwise direction and stop after covering 12 sides, then which of the following statement is true?

- A. Bindiya is to the north of Sanika
- B. Ashini is to the south-west of Bindiya
- C. Ashini is to the east of Sanika
- D. Bindiya is to the north-west of Sanika
- E. Sanika is to the north-east of Ashini

Solution:- B. Ashini is to the south-west of Bindiya

From the positions mentioned in the figure of above question, if all of them run in the anticlockwise direction covering two sides and then stop, then which of the following statements is correct?

- A. Bindiya is to the west of Sanika
- B. Ashini is to the south of Bindiya
- C. Ashini is to the south-west of Sanika
- D. Bindiya is to the north-west of Ashini
- E. Bindiya is to the south of Sanika

Solution:- D. Bindiya is to the north-west of Ashini

DIRECTIONS ( EXAMPLE 23:-) : Read the following information to answer the questions

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given below it.

Pradeep starts walking from his college towards his house. He starts from the front gate of his College and walks 5 km, then turns left and walks 2 km, then turns left again and walks 4 km, then he turns to his right and walks 3 km, then turns left and walks 1 km and then turns to his left again and walks 4 km, then turns to his right and walks 10 km and finally turns right and walks 3 km and thus reaches the front gate of his house.

If Pradeep's house is facing south, in which direction did he start walking?

- A. West
- B. North
- C. South
- D. East
- E. None of these

Solution:- D. East

Pradeep's house is in what direction with respect to his college If Pradeep's house is facing south?

- A. Northwest
- B. Southwest
- C. South
- D. Cannot be determined
- E. None of these

Solution:- A. Northwest

 DIRECTIONS (EXAMPLE 24:-) : Read the following information carefully and answer the questions given below it:

- (A) $P \alpha Q$ means Q is to the right of P at a distance of one metre.
- (B) $P \beta Q$ means Q is to the North of P at a distance of one metre.
- (C) $P \lambda Q$ means Q is to the left of P at a distance of one metre.
- (D) $P \eta Q$ means Q is to the South of P at a distance of one metre.
- (E) In each of the following questions all persons face South.

If $A \eta B \lambda L \beta K$, then K is in which direction with respect to A?

- A. South
- B. East
- C. North
- D. West
- E. None of these

Solution:- B. East

If $G \alpha L \eta R \alpha M$ then M is in which direction with respect to L?

- A. North-east
- B. North-west
- C. South-east
- D. South-west

Analytical Skills-I

E. None of these

Solution:- D. South-west

If $A \propto B \wedge C \beta D$, then D is in which direction with respect to A?

A. North

B. South

C. East

D. North-east

E. None of these

Solution:- A. North

Pran and Khan start from their office and walks in opposite direction, each traveling 10 km. Pran then turns left and walks 10 km. While Khan turns right and walks 10 km. How far they are now from each other?

A. 0 km

B. 5 km

C. 10 km

D. 20 km

E. None of these

Solution:- D. 20 km

Facing to the South Mr. Jha turns in certain ways, which of the following turns will not lead him to the same side?

A. left, left, right, left, left, right

B. left, left, left, right, right, right

C. left, right, left, right, left, right

D. right, left, right, left, right, left

E. None of these

Solution:- A. left, left, right, left, left, right

One morning after sunrise, Vikram and Shailaish were standing in a town with their backs towards each other. Vikram's shadow fell exactly towards left and side. Which direction was Shailaish facing ?

A. East

B. West

C. North

D. South

E. North south

Solution:- D. South

Summary

The key concepts learned from this unit are: -

- We have learnt how to calculate Distance Related Questions
- We have learnt how to calculate Left Right Movement.

Keywords

- Distance
- Directions

Self Assessment

1. If North-east becomes West and South-east becomes North then what will West becomes?
A. South-east
B. North-east
C. South
D. North-west
2. A man walks 6 km towards the north, then turns towards his left and walks for 4 km. He again turns left and walks for 6 km. At this point he turns to his right and walks for 6 km. How many km and in what direction is he from the starting point?
A. 10 km, West
B. 6 km, South
C. 4 km, South
D. 8 km, West
3. A river flows west to east and on the way turns left and goes in a semi-circle round a hillock and then turns left at right angles. In which direction is the river finally flowing?
A. West
B. East
C. North
D. South
4. While facing East you turn to your left and walk 10 yards; then turn to your left and walk 10 yards; and now you turn 45° towards your right and go straight to cover 25 yards. Now, in which direction are you from your starting point ?
A. North-East
B. South-West
C. East
D. North-West
5. A direction pole was situated on the crossing. Due to an accident the pole turned in such a manner that the pointer which was showing East, started showing South. One traveler went to the wrong direction thinking it to be West. In what direction actually he was travelling ?
A. South
B. East
C. West
D. North
6. Five friends are playing a game. Nitin is facing South. He follows instructions given by his friends.
(i) When Kimi touch him, Nitin walks 50 metre.
(ii) When Neha touch him, Nitin turns right and walks 40 metre.
(iii) When Annu touch him, Nitin turns left and walks 80 metre.
(iv) When Punit touch him, Nitin walks 70 metre in the direction opposite to which he faces.
If Kimi, Annu, Punit and Neha touch him in that order, how far does Nitin reach from his starting point?
A. 60 metre.
B. 72 metre.
C. 80 metre.
D. 75 metre.
7. Pradeep starts walking from his college towards his house. He starts from the front gate of his College and walks 5 km, then turns left and walks 2 km, then turns left again and walks 4 km, then he turns to his right and walks 3 km, then turns left and walks 1 km and then turns to his left again and walks 4 km, then turns to his right and walks 10 km and finally turns right and walks 3 km and thus reaches the front gate of his house.

Analytical Skills-I

- If Pradeep's house is facing south, in which direction did he start walking?
- A. West
 - B. North
 - C. South
 - D. East
8. I am facing East. Turning to the right I go 20 m, then turning to the left I go 20 m and turning to the right I go 20 m, then again turning to the right I go 40 m and then again I go 40 m to the right. In which direction am I from my original position?
- A. North
 - B. West
 - C. South
 - D. East
9. Bhairav walked 30 ft towards North, then took a left turn and walked 15 ft. He again took a left turn and walked 30 ft. How far and in which direction is Bhairav from the starting point?
- A. 15 ft to the West
 - B. 45 ft to the South
 - C. 30 ft to the East
 - D. 15 ft to the North
10. Seema started early in the morning on the road towards the Sun. After some time she turned to her left. Again after some time she turned to her right. After moving some distance she again turned to her right and began to move. At this time, in what direction was she moving?
- A. South
 - B. North-West
 - C. North-East
 - D. East
11. At dusk, Rohit started walking facing the sun. After a while, he met his friend and both turned to their left. They halted for a while and started moving by turning again to their right. Finally Rohit waved 'good bye' to his friend and took a left turn at a corner. At which direction is Rohit moving now?
- A. South
 - B. West
 - C. North
 - D. East
12. Sumi ran a distance of 40 m towards South. She then turned to the right and ran for about 15 m, turned right again and ran 50 m. Turning to right then ran for 15 m. Finally she turned to the left an angle of 45° and ran. In which direction was she running finally?
- A. South-East
 - B. South-West
 - C. North-East
 - D. North-West

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13. A man starting from his home moves 4 km towards East, then he turns right and moves 3 km. Now what will be the minimum distance covered by him to come back to his home?
- A. 4
B. 5
C. 12
D. 13

Answers for Self Assessment

1. A 2. A 3. B 4. D 5. A
6. B 7. D 8. B 9. D 10. A
11. A 12. D 13. B

Review Questions

- Satish starts from his house and takes two right turns and then one left turn. Now he is moving towards south. In which direction Satish started from his house?
- A man is facing west. He turns 45° in the clockwise direction and then another 180° in the same direction and then 270° in the anti-clockwise direction. Which direction is he facing now?
- Sumi ran a distance of 40 m towards South. She then turned to the right and ran for about 15 m, turned right again and ran 50 m. Turning to right then ran for 15 m. Finally she turned to the left an angle of 45° and ran. In which direction was she running finally?
- A child is looking for his father. He went 90 metres in the east before turning to his right. He went 20 metres before turning to his right again to look for his father at his uncle's place 30 metres from this point. His father was not there. From there, he went 100 metres to his north before meeting his father in a street. How far did the son meet his father from the starting point?
- Deepa moved a distance of 75 metres towards the north. She then turned to the left and walking for about 25 metres, turned left again and walked 80 metres. Finally, she turned to the right at an angle of 45° . In which direction was she moving finally?
- From his house, Rahul went 25 km to north. Then he turned towards west and covered 15 km. Then he turned south and covered 10 km. Finally, turning to east, he covered 15 km. In which direction is he from his house?
- Rasik walked 20 m towards north. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many metres is he from the starting position?
- One morning after sunrise, Suresh was standing facing a pole. The shadow of the pole fell exactly to his right. To which direction was he facing?
- One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?
- Ranuka started walking from her house, she first walked for 3 km towards west, then she turned towards north and moved 4 km in that direction. How far is renuka from her house?



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle